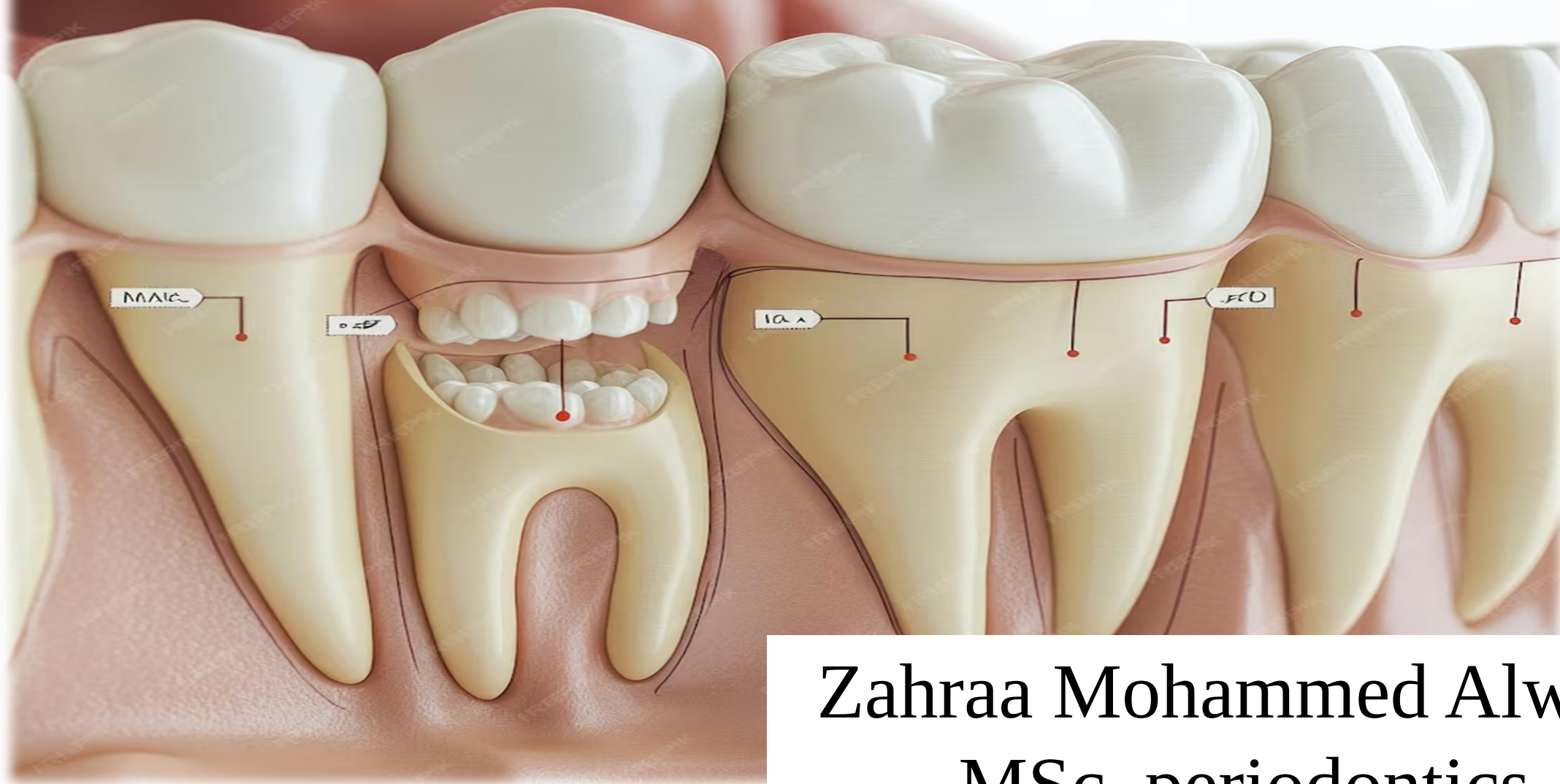


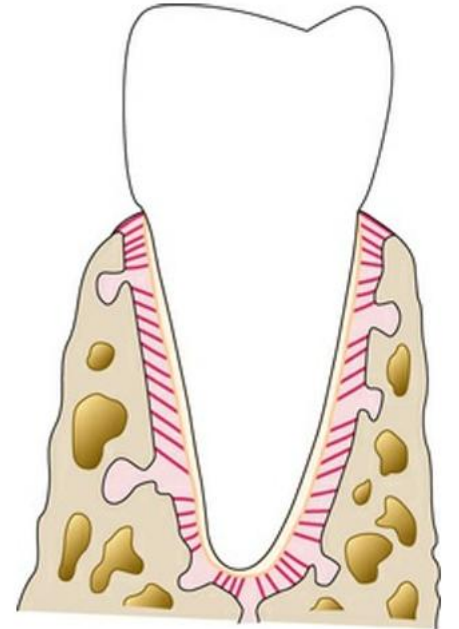
Epidemiology of periodontal disease



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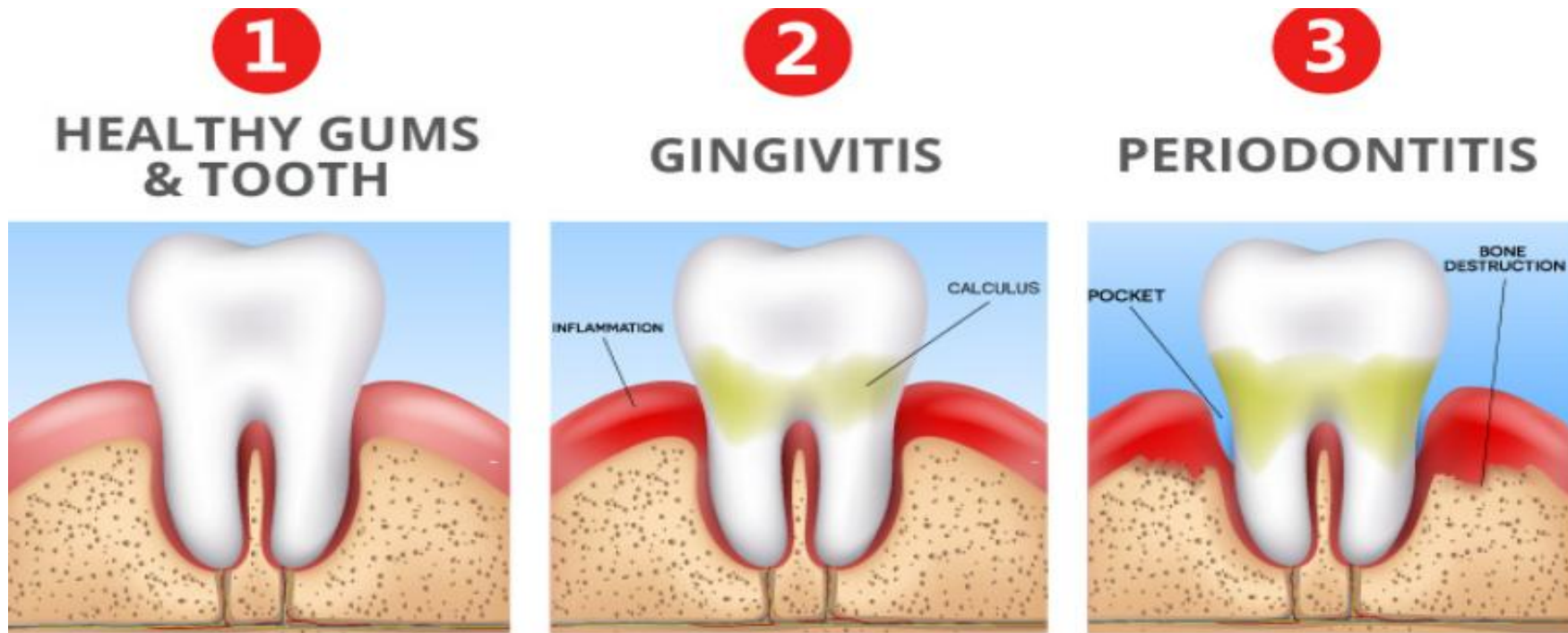
Periodontium consist of

- Gingiva
- Periodontal ligament,
- Cementum
- Alveolar bone.



Periodontal diseases range from gingival inflammation (**gingivitis**) to serious disease (**periodontitis**).

Gingivitis is a mild form of gum disease that can usually be reversed with daily brushing and flossing, and regular cleaning by a dentist or dental hygienist. This form of gum disease does not include any loss of bone. When gingivitis is not treated, it can advance to Periodontitis that result in major damage to the soft tissue and bone that support the teeth.



Etiological factors:

It is well established that the periodontal disease is initiated by **bacterial plaque**, but other etiologic factors exist those which predispose to plaque accumulation and those which modify the inflammatory response.

I. Local factors

A. Deposits on teeth:

1. Dental plaque : soft, non-mineralized, bacterial deposit which forms on teeth
2. Dental calculus: Mineralized dental plaque



B. Abnormal Habits:

1. Unilateral mastication
2. Clenching and bruxism

Clenching: is simply holding the teeth together and tightening the jaw muscles.

Bruxism: is a condition of grinding of teeth as the jaw moves forcefully either from side to side or back and forth. They occur unconsciously during wake up or sleeping

3. Abnormal biting habits

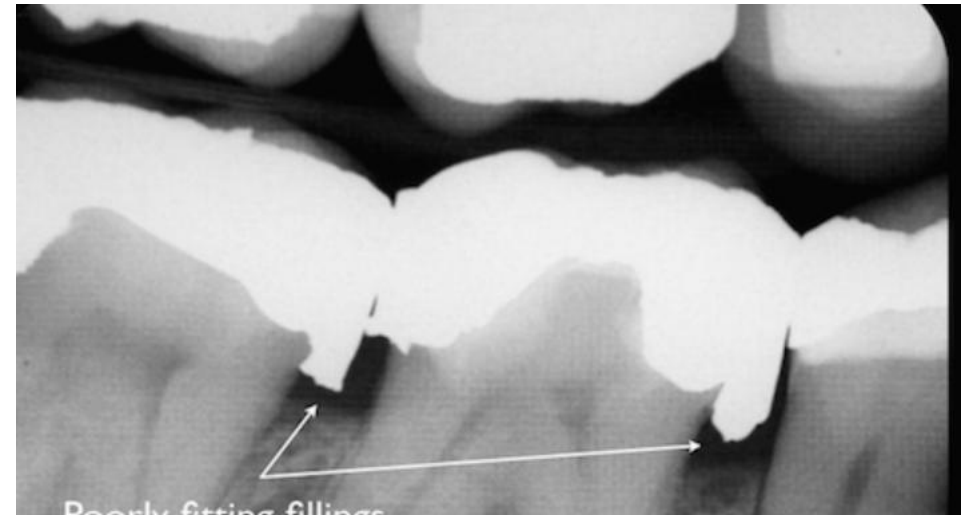


C. Abnormal Anatomy: tooth form and proximal contact

D- Tooth position (crowding): Gingivitis is more common and more severe around malaligned teeth because predisposes to plaque retention and interferes with proper plaque removal.

E-Improper brushing technique: besides resulting in inadequate plaque removal, and can also cause gingival recession.

F. Other irritants : Iatrogenic (Overhang restoration), mouth breather



■ II- Systemic factors

- Malnutrition
- Leukemia
- Diabetes



FIG. 61.8 A 74-year-old patient with acute leukemia. Gingival enlargement is observed in combination with a hemorrhagic appearance, which sometimes causes spontaneous bleeding.



FIG. 61.9 Generalized periodontitis in a patient with poorly controlled type 2



Risk Factors: are variables associated with an increased chance of disease development.

- **Smoking** is one of the major risk factors for periodontitis. Smoking has been shown to affect the vasculature, the humoral and cellular immune responses, cell signaling processes, and tissue homeostasis. Moreover, it affects bacterial colonization and aggregation, and results in higher levels of colonization by key periodontal pathogens.



- **Hormonal changes in girls/women.** These changes can make gums more sensitive and make it easier for gingivitis to develop. gingivitis is seen at the time of menstruation and pregnancy
- **Oral contraceptive:** studies report increased gingival inflammation and increased periodontal destruction.
- **Medications**
Drugs that may cause gingival overgrowth include anticonvulsant (e.g. phenytoin), immunosuppressant (e.g. cyclosporine A), and calcium channel blockers (e.g. nifedipine, verapamil).
- **Race/ethnicity:** Differences in the prevalence of periodontitis between countries and different races have been demonstrated.



- **Obesity:** research has shown that obesity may increase the risk of periodontal disease.
- **Socio-economic Status :**High income group have lower periodontal disease rate than the lower income group, probably because they have the means and can afford dental treatment.
- **Restoration:** Bacteria accumulate more readily on filled surface, than on tooth surface. A smooth and highly polished filling is easier to clean than a rough surface and hence there is lesser degree of plaque accumulation.

Epidemiology of periodontal disease:

1. Interproximal areas are the most severely affected by gingivitis followed by buccal and lingual surfaces.
2. Gingivitis is more severe in the upper arch than lower arch for interproximal and buccal areas. For the lingual areas, gingivitis is found to be more severe in the lower arch.
3. The most severely affected teeth by periodontal disease are upper molars and lower incisors whereas the least affected are lower premolars and upper canines.

4. Higher tendency toward gingivitis is on **the right half** of the area than on the left half. This may be because of difficulty that right-handed person has in brushing the right half of the mouth

5. Gingivitis has been observed in children younger than 5 years of age .In general prevalence and severity of gingivitis increase with the age ,beginning at approximately 5 years of age ,reaching their highest point in puberty and then very gradually decreasing but remaining relatively high throughout life.

6-gender: Periodontal disease is **more common in males**. The reasons for these gender differences are not clear, but it is thought to be related to poorer oral hygiene level, which is usually observed among males

Elements of prevention of periodontal disease:

- a) Oral health education for individuals and community
- b) Tooth brushing
- c) Oral hygiene aids
- d) Chemotherapy of dental plaque infection (Chlorhexidin)
- e) Dietary factors in plaque control
- f) Disease control by professionals

